

Roma Solapurkar

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EDUCATION

Johns Hopkins University

M.S. in Artificial Intelligence

Baltimore, MD | 2026 – 2029

University of California, Santa Cruz

B.S. in Computer Science | GPA: 3.85/4.0, Highest Honors

March 2025

Relevant Coursework: Algorithms, Data Structures, Operating Systems, Computer Systems Design

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript, SQL, HTML, CSS

Data & ML: NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn, PyTorch

Frameworks & Tools: Spring Boot, REST APIs, Apache NiFi, RabbitMQ, Docker, Kubernetes, Jenkins, Git

Platforms: ServiceNow, Firebase, py4web, Grafana, Oracle MySQL

Concepts: Microservice architecture, REST API design, asynchronous messaging, message queuing, data pipelines, relational databases, object-oriented design, distributed systems, observability, unit & integration testing, code review

CERTIFICATIONS

CompTIA Security+ (2025) | Machine Learning Foundations – Cornell Tech (2024) | Introduction to AI Agents – DataCamp (2026)

Active U.S. Security Clearance | Level available upon request

EXPERIENCE

Associate Software Engineer

June 2025 – Present

L3Harris Technologies

- Built and extended 6+ Java (Spring Boot) microservices and plugins, designing REST API endpoints to define message routes and using RabbitMQ to publish and consume messages across services; leveraged Apache NiFi to route and transform messages to different destinations.
- Collaborated on a cross-team plugin integration applying object-oriented design principles (layered architecture, separation of concerns); implemented message parsing logic and wrote data to an Oracle MySQL database, coordinating design and integration across two engineering teams through to deployment.
- Deployed services to a Kubernetes-clustered environment; published and consumed messages via RabbitMQ, and used Apache NiFi to route and transform messages across service destinations asynchronously, supporting decoupled inter-service communication.
- Standardized local development environments using Docker with mock databases, replacing ad hoc per-developer setups to support consistent unit and integration testing in isolation before staging deployment.
- Built, tested, and deployed services using Jenkins CI/CD pipelines following Agile development practices.
- Participate in code review as both author and reviewer, providing written feedback on peers' pull requests to maintain code quality and shared engineering standards.
- Used the Grafana API to programmatically collect platform metrics and route them to a destination, improving system observability across 6+ production services.
- Presented technical updates and team deliverables directly to U.S. government and Space Force stakeholders, communicating engineering progress and driving alignment on project status.
- Independently designing and building an AI research agent using the ReAct (Reasoning + Acting) framework via the OpenAI API; architected the prompt structure, tool interface, and reasoning loop so the agent can complete multi-step research tasks end-to-end without human re-prompting.

Software Engineer Intern

June – September 2024

Asurion

- Overhauled ServiceNow platform stability by repairing broken forms and implementing CI practices, resulting in a 50% reduction in manual workflow overhead for the call-center operations team.
- Built a custom ServiceNow application integrating the Observe.AI API to automate agent performance analysis, replacing a manual review process and closing the feedback loop from days to near-real-time.
- Won the internal innovation challenge audience vote for a solution developed through independent root-cause analysis and original system design.

Breakthrough Tech AI/ML Fellow

May 2024 – January 2025

Cornell Tech & Google

- Built an ML model using NLP and deep learning on Google Trends data to predict search query rankings; collaborated with Google Data Scientists and MLEs on feature selection and model evaluation.

- Analyzed the Airbnb NYC dataset using decision tree and KNN models, iterating on hyperparameter tuning and feature engineering to improve classification accuracy.
- Worked in cross-functional teams with direct mentorship from Google Data Scientists and Machine Learning Engineers.

PROJECTS

AI Research Agent (ReAct Framework)

2025 – Present

- Solo-designed and building an AI research agent using the ReAct prompting pattern; architected the reasoning loop, tool interface, and failure-handling logic so the agent completes multi-step research tasks end-to-end and recovers from bad tool calls without human re-prompting.

Task Management Web Application

March – June 2024

- Built a full-stack task management app in py4web, Python, and JavaScript with org-aware role-based access control, task lifecycle state machines, and a hierarchical manager approval chain.

PostNote Web Application

September – December 2023

- Developed a full-stack note-sharing app with course integration, file upload/download, and real-time peer collaboration using Dart, Flutter, and Firebase.

LEADERSHIP

Space Ambassador

August 2025 – Present

Space Foundation

- Deliver hands-on STEM and aerospace instruction in Discovery Center labs; judged the National Space Sustainability Competition; recognized as New Volunteer of the Month.
- Coordinated cross-organizational engagement between L3Harris Technologies and the Space Foundation during Engineers Week, connecting industry professionals with STEM education initiatives.

Solution Challenge Coordinator

December 2021 – March 2025

UCSC Google Developer Student Club

- Organized and managed 25 team projects for the Solution Challenge; led technical workshops and internal operations.